Shayaan Rashedin

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linkedin.com/in/shayaanrashedin | github.com/ShayaanRashedin | Portfolio: shayaanrashedin.github.io

Skills

Technical Skills: C++, C, HCS12 Assembly, Python, Bash, Linux, Windows, POSIX, TCP/IP, sockets, epoll, multithreading, procfs, embedded firmware, GPIO, ADC, interrupts, SQLite, SQL, REST APIs, JavaScript, CMake, CTest, Git, GitHub Actions, Docker, Splunk, Wireshark, Nmap

Education

Bachelor of Engineering - Computer Engineering *September 2023 - Expected April 2027*

Toronto Metropolitan University, Toronto, ON

- Relevant Coursework: Operating Systems; Computer Networks; Network Security; Embedded Systems Design; Advanced Algorithms; Computer Organization and Architecture; Digital Systems Engineering; Systems-on-Chip Design

Projects

SyncWire | C++20 Linux File Transfer and Synchronization

Completed personal project | github.com/ShayaanRashedin/syncwire

- Designed a versioned binary TCP protocol with incremental framing, network byte order, bounded payloads, partial I/O handling, and mutual HMAC-SHA256 authentication.
- Built an epoll accept loop with a bounded queue and std::jthread worker pool, plus resumable uploads, CRC-32 verification, fsync, and atomic file replacement.
- Verified protocol, filesystem, concurrency, restart recovery, and real TCP behavior with CTest, integration tests, ASan/UBSan, and GitHub Actions.

NetWatch | C++20 Linux Network Observability

Completed personal project | github.com/ShayaanRashedin/netwatch

- Parsed Linux procfs TCP/UDP tables across IPv4/IPv6 and correlated socket inodes with PIDs, users, executable paths, command lines, and process start times.
- Implemented rule-based behavioral detection and a bounded producer-consumer pipeline for transactional SQLite event and alert persistence.
- Delivered a REST/JSON API, JavaScript dashboard, Catch2/CTest suite, and Docker, systemd, CPack, and GitHub Actions workflows.

EEBOT Robot Guidance | HCS12 Assembly Embedded Firmware

Three-person COE538 project | github.com/ShayaanRashedin/eebot-robot-guidance

- Integrated five infrared channels, ADC, GPIO motor control, bumper inputs, timer interrupts, and a 20x2 LCD through register-level MC9S12C32 firmware.
- Implemented a 9-state navigation controller with calibrated tape following, junction handling, fault timeouts, and fixed 32-entry route memory for backtracking and return traversal.

Network Threat Detector | Python Detection Pipeline

Completed personal project | github.com/ShayaanRashedin/network-threat-detector

- Built a modular Python network-log detection pipeline with four explainable rules, SQLite evidence persistence, and automated Airflow execution through Docker Compose.

Cybersecurity Homelab | Virtualized Detection and Vulnerability Assessment

Documented lab | shayaanrashedin.github.io/homelab.html

- Built an isolated Kali, Metasploitable2, and Splunk lab for reconnaissance, vulnerability assessment, controlled validation, and SPL-based brute-force detection mapped to MITRE ATT&CK.

Certifications

Google Cybersecurity Professional Certificate *June 2025*

Google IT Support Professional Certificate *February 2025*

Experience

VP of Education - Toronto Met Cyber Security Association, Toronto, ON *October 2025 - Present*

- Plan workshops and hands-on labs, create technical learning materials, and facilitate peer learning in cybersecurity, Linux, networking, and technical career exploration.

Youth Engineering Instructor - Toronto High Park Young Engineers, Toronto, ON *September 2024 - September 2026*

- Delivered LEGO-based engineering lessons to groups of 20+ students, explaining transmissions, gear ratios, and basic programming logic.
- Led hands-on builds and design challenges that developed teamwork, problem-solving, and creative thinking while making STEM concepts accessible.